

Worksheet 5 *Power functions*Prepares you for lab manual pages **Exp5, Exp6, Exp7, Exp8** (section L2 S4)

Name:

Name of TA:

1. Suppose that $F(t) = 9t^{2/3}$.(a) Draw a time series for the quantity F . (Use Desmos to help if you wish.)(b) What is $F(8)$? (Give the exact value; it is a whole number.)(c) What is $F(8.01)$? (Use a calculator.)(d) What is ΔF on the time interval from $t = 8$ to $t = 8.01$?(e) Estimate dF/dt when $t = 8$.**2.** A circular spot of mold is growing on the ceiling. Its radius is $R(t) = 2t$, where t is time in days and R is in millimeters.

(a) Describe the area of the mold spot as a function of time.

 $A(t) =$ (b) Estimate dA/dt when $t = 3$. $dA/dt \approx$ **3.** Carefully draw a tangent line with a ruler, and use it to estimate dY/dt when $t = 3$. $dY/dt =$ $\text{ when } t = 3.$